Real-Time Multilingual Voice Interpreter System

Project Goal:

Two people should be able to talk in their native language and hear each other's speech translated and spoken out loud in their own language — via a website or physical device.

A. Web App Version (React + Node.js + APIs)

Tech Stack:

Frontend: React + Web Audio API

Backend: Node.js / Express

STT: OpenAI Whisper API / Google Speech

Translation: Google Translate API / LibreTranslate

TTS: Google TTS / Coqui / ElevenLabs

Hosting: Vercel / Netlify (frontend), Render / Railway (backend)

Features:

1. UI Layout (React)
   - Two panels for users
   - Microphone buttons, language selectors, speaker toggle
   - Live chat translation log

2. Audio Capture (Web Audio API)
   - Capture mic input from browser
   - Send audio blob to backend

3. Speech-to-Text (STT)

- Use Whisper or Google Speech to convert audio to text

4. Translation
- Translate source text using Google Translate API

5. Text-to-Speech (TTS)
- Speak out loud using TTS API in target language

6. Full Cycle: Speak → Text → Translate → Speak

B. Hardware Version (Raspberry Pi)
Components:
- Raspberry Pi 4/Zero
- USB Mic + Speaker
- Optional Touchscreen
- Wi-Fi Internet
- Power Supply

Software:
- OS: Raspbian OS
- STT: Whisper via Python
- Translation: Google Translate API
- TTS: Coqui / pyttsx3 / gTTS
- Interface: Python Tkinter / Flask

Flow:
1. Person speaks → Raspberry Pi captures audio

2. Converts to text → Translates → Speaks in target language

3. Two-way cycle for conversation

Folder Structure:

```
language-interpreter/
├── web-app/
│   ├── client/ (React frontend)
│   └── server/ (Node.js backend)
├── hardware-device/
│   ├── main.py
│   ├── audio_utils.py
│   ├── tts_module.py
│   ├── stt_module.py
│   └── translate_module.py
└── README.md
```

Free and Paid APIs:

STT: Whisper (free), Google Speech (paid)

Translation: LibreTranslate (free), Google Translate (paid)

TTS: Coqui/pyttsx3 (free), ElevenLabs (paid)